Math 133: Business Optimization

Instructor Key

Suggested pacing: 3, 5, 7, 7, and 8 minutes for Problems 1–5, respectively.

1. $P'(x) = -2x + 24$, so the only critical value is $x = 12$. The derivative is positive before 12 and negative after 12. Also,

$$P(0) = -44, \qquad P(12) = 100, \qquad P(20) = 36.$$

The company should produce **12,000 computers** per month for a maximum monthly profit of **$100,000**. Both x and $P(x)$ are measured in thousands.

2. Revenue and cost are in dollars:

$$R(q) = q(14 - 0.02q), \qquad C(q) = 100 + 2q.$$

Thus the daily profit is

$$P(q) = -0.02q^2 + 12q - 100, \qquad P'(q) = -0.04q + 12.$$

The derivative vanishes at $q = 300$, which is outside the feasible interval $[0, 250]$. Since $P'(q) > 0$ throughout this interval, profit is increasing and its maximum occurs at the upper boundary, $q = 250$. The endpoint values are

$$P(0) = -100, \qquad P(250) = 1650.$$

At $q = 250$, the price is $p(250) = 14 - 0.02(250) = 9$. Charge **$9 per mug**, sell **250 mugs**, and earn a maximum daily profit of **$1,650**. The capacity constraint is binding.

3. Differentiate and factor:

$$P'(x) = -3x^2 + 18x - 15 = -3(x - 1)(x - 5).$$

Both critical values, $x = 1$ and $x = 5$, lie in $(0, 8)$.

x	$(0,1)$	1	$(1,5)$	5	$(5,8)$
$P'(x)$	$-$	0	$+$	0	$-$

At $x = 1$ there is a local minimum, $P(1) = -17$ thousand dollars. At $x = 5$ there is a local maximum, $P(5) = 15$ thousand dollars. For the absolute maximum, compare both critical values and the endpoints:

x	0	1	5	8
$P(x)$ (thousand dollars)	-10	-17	15	-66

Produce **5,000 units** per week for a maximum weekly profit of **$15,000**. The first critical value is a local minimum and is not an optimal production choice.

4. Because x is in thousands and the unit price is in dollars, revenue in thousands of dollars is

$$R(x) = xp(x) = 18xe^{-x/3}.$$

Using the product and chain rules,

$$R'(x) = 18e^{-x/3} - 6xe^{-x/3} = 6e^{-x/3}(3 - x).$$

The exponential factor is always positive. Thus the only critical value is $x = 3$, and R' changes from positive to negative there. The candidate revenues, in thousands of dollars, are

$$R(0) = 0, \qquad R(3) = \frac{54}{e} \approx 19.86549, \qquad R(9) = \frac{162}{e^3} \approx 8.06551.$$

The revenue-maximizing price is

$$p(3) = \frac{18}{e} \approx \boxed{\$6.62 \text{ per unit}}.$$

The corresponding quantity is **3,000 units per week**. Maximum weekly revenue is

$$\frac{54{,}000}{e} \text{ dollars} \approx \boxed{\$19{,}865.49}.$$

5. Revenue in thousands of dollars is $R(x) = x(6 - \ln x)$. Since x thousand hot dogs cost x thousand dollars to make, the cost function in thousands of dollars is $C(x) = x + 5$. Hence

$$P(x) = R(x) - C(x) = x(5 - \ln x) - 5.$$

The product rule gives

$$P'(x) = (5 - \ln x) - 1 = 4 - \ln x.$$

The only critical value is $x = e^4 \approx 54.59815$, which lies in $[1, 60]$. Profit increases for $x < e^4$ and decreases for $x > e^4$. Comparing candidates in thousands of dollars,

$$P(1) = 0, \qquad P(e^4) = e^4 - 5 \approx 49.59815, \qquad P(60) = 295 - 60\ln 60 \approx 49.33933.$$

The optimal price is $p(e^4) = 6 - 4 = \boxed{\$2 \text{ per hot dog}}$. Sales are $1000e^4 \approx \boxed{54{,}598 \text{ hot dogs per game}}$. Maximum profit is

$$1000(e^4 - 5) \text{ dollars} \approx \boxed{\$49{,}598.15 \text{ per game}}.$$

Fixed costs affect the profit level but not the maximizing price or quantity in this model.